



Data Science Foundations

Outline

- Introduction to Data Science
- Roles
- Lifecycle
- Reproducible Workflows

Acknowledgement : some slides are adapted from Lectures of <https://ds100.org/>

What Is Data Science? 🔍

- **Definition:** collecting, analyzing, and modeling data to **generate insights** that drive **decision-making**

 **Data** →  **Insight** →  **Decision**

 **The goal:** better decisions, not pretty charts

- Learning about the world from data using computation
- Not just reporting: asks new questions, not just old ones
- **Interdisciplinary:** math, programming, machine learning, domain know-how
- **Structured and unstructured data:** tables, text, images, audio, sensors data.

Difference Between Data, Information, and Insights

Level	Example
Data	Sales = \$100,000 in October, \$130,000 in November
Information	Sales increased by 30% from October to November
Insight	Holiday shopping behavior drives demand, creating an opportunity for seasonal promotions

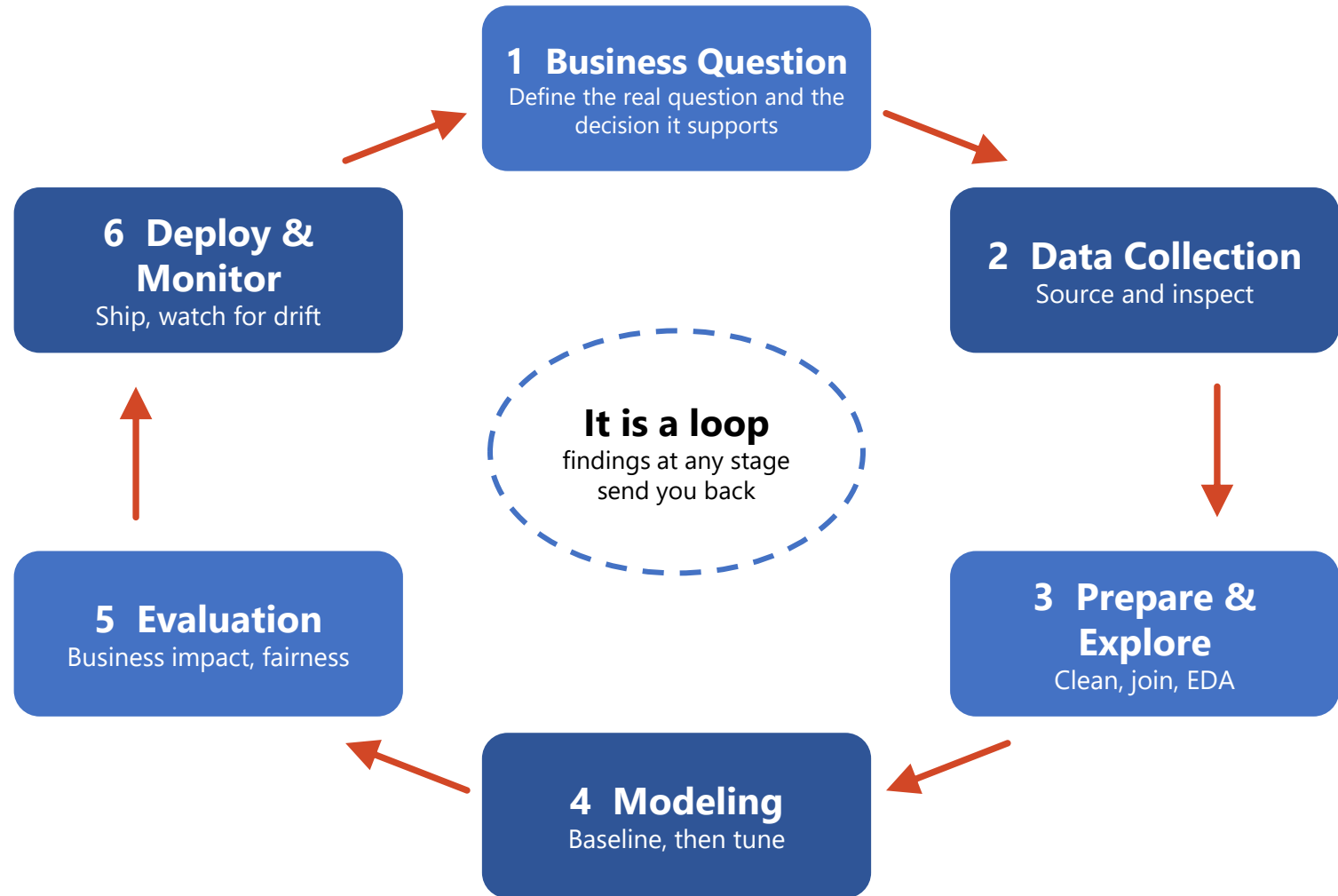
A good insight is:

- **Relevant** to the business or research question
- **Actionable** and useful for decision-making
- **Evidence-based** and supported by data
- **Non-obvious**, revealing something not immediately apparent
- **Impactful**, leading to improved outcomes

Where Data Science Creates Value

-  **Healthcare:** imaging diagnosis, readmission risk, drug discovery
-  **Finance:** fraud detection, credit scoring, risk management
-  **Retail:** demand forecasting, recommenders, dynamic pricing
-  **Logistics:** route optimization, predictive maintenance, inventory
-  **Education:** admissions analytics, early warning for at-risk students
-  **Energy:** demand forecasting, policy evaluation

Data Science Lifecycle



Types of Data Analysis

-  **Descriptive** — “What happened?” Dashboards and KPIs.

Example: applications, enrollment, and pass rates by program each term.

-  **Diagnostic** — “Why did it happen?” Identifies the factors and root causes behind observed outcomes through drill-down and comparative analysis.

Example: Enrollment decreased by 15%. Diagnostic analysis: decline was driven by lower admissions in CE and a reduction in international student applications.

-  **Predictive** — “What is likely to happen?” Forecasting and machine learning.

Example: flagging first-year students at risk of failing a course.

-  **Prescriptive** — “What should we do?” Optimization and recommendation.

Example: which recruitment visits and scholarship offers to prioritize.

-  **Cognitive** — “Can the system reason like a human?” LLMs and NLP over unstructured data. *Example:* an AI tutor giving feedback on student essays.

 **Rising difficulty, rising value:** each level builds on the one before it.

What we will focus on in this course

- **Foundations — getting trustworthy data**
 - Problem formulation
 - Data acquisition, quality, and representativeness
 - Data wrangling: cleaning, transformation, integration, feature engineering
- **Exploration — finding patterns in data**
 - Exploratory data analysis and visualization
- **Inference — drawing reliable conclusions**
 - Probability: sampling and uncertainty
 - Statistical testing, estimation, and A/B testing
 - Causal inference and experimental design
- **Prediction — making informed guesses**
 - Regression and classification
 - Clustering and dimensionality reduction

Who Does What: Data and AI Roles

-  **Data Engineer** — builds and maintains pipelines, warehouses, and lakes so data arrives clean, on time, and at scale.
 - SQL, Spark, ETL/ELT, cloud
-  **Data Analyst** — answers defined business questions from existing data; dashboards, KPIs, and reporting.
 - SQL, Excel, BI tools, visualization.
-  **Data Scientist** — frames the question, experiments, and models; statistics, causal thinking, and prototype ML.
 - Python/R, statistics, A/B testing.
-  **ML Engineer** — takes models to production and keeps them healthy: deployment, scaling, monitoring, retraining.
 - MLOps, APIs, containers.
-  **AI Engineer** — builds applications on foundation models and LLMs: prompting, retrieval, agents, evaluation, and guardrails.

 **In practice:** titles overlap and small teams blend them

Roles at a Glance: Question, Output, Tools

 **Data Engineer** — Question: “Is the data available and trustworthy?”

Output: reliable pipelines and tables.

Tools: SQL, Spark, Airflow, cloud storage.

 **Data Analyst** — Question: “What happened, and what does it mean for the business?”

Output: dashboards, reports, recommendations.

Tools: SQL, BI tools, visualization.

 **Data Scientist** — Question: “Why did it happen, and what will happen next?”

Output: experiments, statistical and ML models.

Tools: Python/R, statistics, experimentation.

 **ML Engineer** — Question: “Will this model work reliably at scale?”

Output: deployed, monitored services.

Tools: MLOps, Docker, CI/CD, cloud.

 **AI Engineer** — Question: “How do we build a product on top of foundation models?”

Output: LLM-powered applications.

Tools: prompting, RAG, agents, evaluation.